

CS471 – Block 2 Worksheet

Lessons 11-15: Markov Decision Processes and Reinforcement Learning

Block 2 Equations Reference

- **Discounted Return:** *Computes the cumulative reward from a sequence of steps, where the discount factor $\gamma \in [0, 1]$ bounds the value of infinite future returns.*

$$G = r_1 + \gamma r_2 + \gamma^2 r_3 + \dots$$

- **Value Iteration (Bellman Update):** *Calculates the maximum expected value of state s by checking all actions a and summing over all possible next states s' , weighted by transition probability $P(s'|s, a)$ and discount factor γ .*

$$V_{k+1}(s) = \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V_k(s')]$$

- **Policy Evaluation:** *Computes the expected value of state s when strictly following a specific, fixed policy π .*

$$V^\pi(s) = \sum_{s'} P(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma V^\pi(s')]$$

- **Q-Learning Update:** *Updates the value of taking action a in state s based on the learning rate α , immediate reward r , and the discounted max expected value of the resulting next state s' .*

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right]$$

Part I: Guided Practice

Lesson 11. MDP Formulation and Markov Property

An autonomous cyber-defense agent routes analyzed data packets between secured enclave servers. A student proposes the following MDP:

- **State:** the packet's current server node.
 - **Actions:** forward, inspect, drop, or encrypt.
 - **Transition model:** a forward action always successfully reaches the intended neighboring server.
 - **Reward:** +5 when a packet reaches the destination enclave; 0 otherwise.
- (a) Identify two pieces of information that should be added to the state to better satisfy the Markov property. Briefly explain why each matters.
- (b) Give one realistic reason a forward action could have more than one outcome. How should the transition model represent this uncertainty?

Lesson 12. Policies, Values, and Return

An autonomous surveillance UAV follows the fixed policy shown below. Assume transitions are deterministic.

State	Policy Action	Immediate Reward	Next State
S_1	Scan	+2	S_2
S_2	Transmit	+6	Terminal

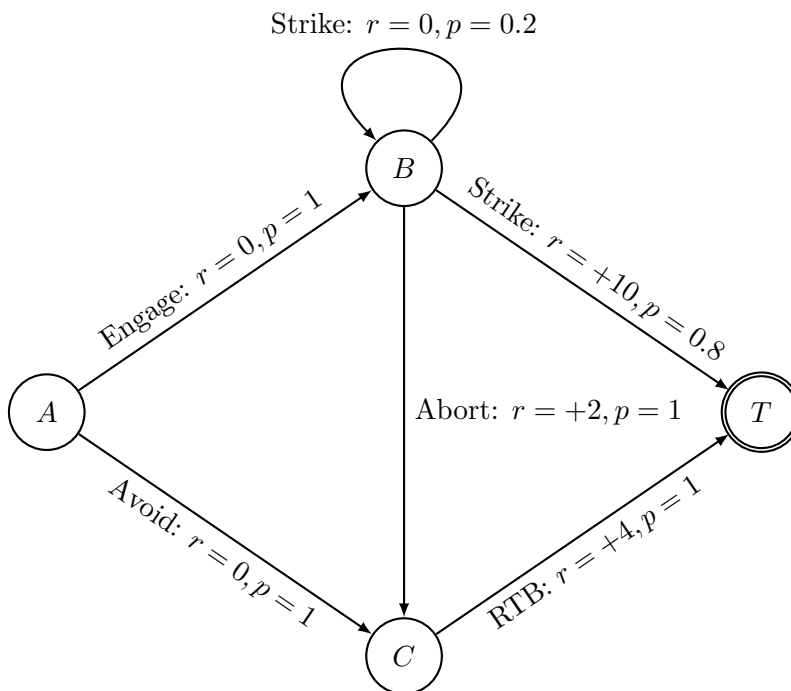
Use a discount factor of $\gamma = 0.5$.

- (a) Starting in state S_1 and following the policy, compute the discounted return.

- (b) What does the state value $V^\pi(S_1)$ represent in this situation?
- (c) Suppose a different policy chooses “Loiter” at S_1 , receiving an immediate reward of +4 and terminating. Which policy gives the larger expected return from S_1 ?

Lesson 13: Value Iteration

An autonomous surface vessel uses the MDP below. Let $\gamma = 0.5$, $V(\text{Terminal}) = 0$, and $V_0(A) = V_0(B) = V_0(C) = 0$. Use synchronous updates.



- (a) Complete the first Value Iteration sweep to find $V_1(A)$, $V_1(B)$, and $V_1(C)$.
- (b) Using V_1 , compute only $V_2(A)$. Show the comparison between the available actions.

- (c) After additional sweeps, suppose the values are: $V_k(A) = 3.80$, $V_k(B) = 8.12$, $V_k(C) = 4.00$, and $V_{k+1}(A) = 3.82$, $V_{k+1}(B) = 8.15$, $V_{k+1}(C) = 4.00$.
- (i) Using the V_{k+1} values, determine the greedy policy action at state A . Show the action comparison.
- (ii) If the convergence tolerance is $\theta = 0.05$, has Value Iteration converged? Briefly justify.

Lesson 14: Policy Iteration

An autonomous logistics agent is being optimized using Policy Iteration. For the current policy, $V^\pi(Y) = 8$ and $V^\pi(Z) = -4$. Use a discount factor of $\gamma = 0.5$. The relevant transitions are:

State	Action	Next State	Reward
X	Option 1	Y	-2
X	Option 2	Z	-2
W	Option 3	Y	-2
W	Option 4	Z	-2

The current policy selects **Option 2** at X and **Option 4** at W .

- (a) Under the current policy, compute $V^\pi(X)$.
- (b) Perform policy improvement.
- (i) At state X , compare the available actions and give $\pi'(X)$.
- (ii) At state W , compare the available actions and give $\pi'(W)$.

- (c) Based on the improved policy, what should Policy Iteration do next? Briefly justify your answer.

Lesson 15: Q-Learning from Experience

A network-security AI has the following Q-values:

$Q(N_1, \text{Scan}) = 0$, $Q(N_2, \text{Patch}) = 0$, $Q(N_2, \text{Ignore}) = 2$.

Use $\alpha = 0.5$ and $\gamma = 0.5$. State T is terminal.

Process the following experiences in order. Use any updated Q-values in the next experience.

- (a) The agent experiences $(N_2, \text{Patch}, 8, T)$. Compute the updated value of $Q(N_2, \text{Patch})$.
- (b) The agent then experiences $(N_1, \text{Scan}, 0, N_2)$. Compute the updated value of $Q(N_1, \text{Scan})$.
- (c) Suppose the experience in part (b) had occurred before the experience in part (a). Would the resulting update to $Q(N_1, \text{Scan})$ be smaller, the same, or larger? Briefly justify using the Q-values at state N_2 .

Part II: Independent Student Practice

Practice 1: MDP Formulation and Markov Property

An orbital satellite navigates between orbital slots to optimize communication coverage.

- **State:** The satellite's current orbital slot.
 - **Actions:** Thrust-Prograde, Thrust-Retrograde, or Sleep.
 - **Transition model:** A Thrust action always successfully moves the satellite to the adjacent slot.
 - **Reward:** +10 when reaching the target comms slot; 0 otherwise.
- (a) Identify two physical or system variables that should be added to the state to better satisfy the Markov property. Briefly explain why each matters.
- (b) Give one realistic reason a Thrust action could have more than one outcome. How should the transition model adapt to this?

Practice 2: Policies, Values, and Return

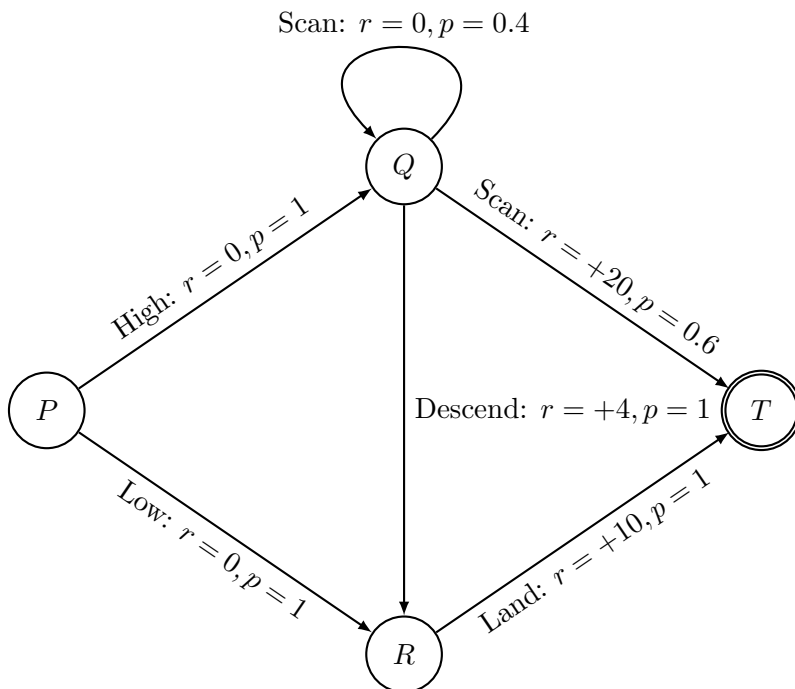
An automated supply convoy follows the deterministic fixed policy below. Use a discount factor of $\gamma = 0.25$.

State	Policy Action	Immediate Reward	Next State
S_A	Drive	+3	S_B
S_B	Unload	+8	Terminal

- (a) Starting in state S_A and following the policy, compute the discounted return.
- (b) Suppose a different policy chooses “Airdrop” at S_A , receiving an immediate reward of +6 and terminating. Which policy gives the larger expected return from S_A ?

Practice 3: Value Iteration

An autonomous UAV navigates mountainous terrain. Let $\gamma = 0.5$, $V(\text{Terminal}) = 0$, and $V_0(P) = V_0(Q) = V_0(R) = 0$. Use synchronous updates.



- (a) Complete the first Value Iteration sweep to find $V_1(P)$, $V_1(Q)$, and $V_1(R)$.
- (b) Using V_1 , compute only $V_2(P)$. Show the comparison between the available actions.
- (c) If the convergence tolerance is $\theta = 0.1$, and the agent calculates $V_k(P) = 5.20$ and $V_{k+1}(P) = 5.26$ (with no other states changing), has Value Iteration converged?

Practice 4: Policy Iteration

An Electronic Warfare AI selects jamming frequencies. For the current policy, $V^\pi(S_C) = 12$ and $V^\pi(S_D) = -8$. Use a discount factor of $\gamma = 0.5$.

State	Action	Next State	Reward
S_A	Act 1	S_C	-1
S_A	Act 2	S_D	-1
S_B	Act 3	S_C	-1
S_B	Act 4	S_D	-1

The current policy selects **Act 2** at S_A and **Act 4** at S_B .

- (a) Under the current policy, compute $V^\pi(S_A)$.
- (b) Perform policy improvement at state S_A . Compare the available actions and give $\pi'(S_A)$.

Practice 5: Q-Learning from Experience

An autonomous rover has the following Q-values: $Q(R_1, \text{Drive}) = 0$, $Q(R_2, \text{Sample}) = 0$, $Q(R_2, \text{Wait}) = 4$. Let $\alpha = 0.5$, $\gamma = 0.5$, and let T be terminal.

- (a) The agent experiences $(R_2, \text{Sample}, 10, T)$. Compute the updated value of $Q(R_2, \text{Sample})$.
- (b) The agent immediately experiences $(R_1, \text{Drive}, 0, R_2)$. Compute the updated value of $Q(R_1, \text{Drive})$.